Integrating GitLab with Moodle

Bachelor Thesis — HEIG-VD

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A unified workflow for programming assignments in computer science education

Platforms Involved

Learning Management Systems (LMS)

- Centralizes course organization and documents
- Manages assignments, submissions, and grading
- Provides a structured environment for teaching and learning

Version Control Systems (VCS)

- Manages source code and development workflows
- Enables collaboration through Git repositories
- Provides project management / DevOps tools

Moodle

GitLab

Context

Moodle and GitLab are both widely used in computer science education, but operate independently leading to fragmented workflows.

Teachers

- Create repositories manually
- Configure GitLab permissions by hand
- Manage multiple platforms

Students

- Inconsistent workflows between courses
- Switch between LMS and VCS platforms
- Grades, deadlines, repositories in different places

Specification

Design a solution around 3 core principles

Automation

- Automate GitLab resource management
- Favor flexibility over rigid workflows

LMS-native

- Provide a unified workflow
- Central access point over fragmented platforms

Extensible Design

- Open-source allowing contributions
- Modular architecture for futur extensions

Proprietary Solutions Risks

> On Friday August 28th GitHub Classroom will fully transition to partner solutions, [...] and the GitHub Classroom website will be decommissioned. [...] We know that change is not easy and many of you have been relying on GitHub Classroom for years [...].

The GitHub Education Team

- No control over platform evolution
- Risk of service discontinuation
- Potential pricing changes
- Limited adaptability to specific needs
- Vendor lock-in

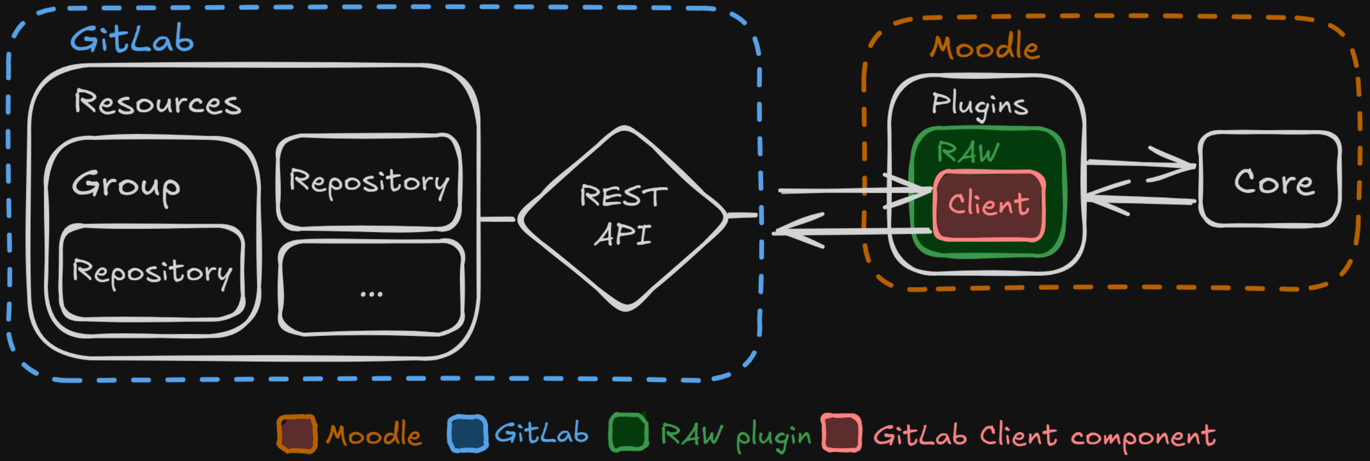
State of the Art

	Automation	Open source	LMS integration	VCS independent
GitHub Classroom	✅	🟡	❌	❌
Codio	✅	❌	🟡	❌
Classroom 50	✅	✅	❌	❌
RepoBee	🟡	✅	❌	✅
Classmoji	✅	✅	✅	🟡
This work	✅	✅	✅	🟡

Solution

Repository Assignment Workflow (RAW)

A Moodle-native, extensible, and open-source solution.



- Follows the 3 design principles defined previously
- Built using two Moodle plugins: **Custom Field**, **Activity Module**

Demonstration

Features

Assignment Management

- RAW activity configuration
  - Due dates
  - Group capacity
  - Reviewers list
- Moodle calendar events and notifications
- Release solution access

User Workflows

- Teacher dashboard
  - Manage groups and members
  - Access template repository
  - Retrieve submissions
- Student interface
  - Create or join groups
  - Access repository information

Features

GitLab Integration

- Automated repository provisioning
- Repository access and permission management
- Webhook synchronization
- Protected files verification
- Custom GitLab host support

Template repository

- Instructions issue
- Solution branch
- Group submissions merge request

Group repository

- Fork of the template repository
- Instructions issue
- Submission merge request

Limitations and Future Work

Limitations

GitLab permission model limits control

- Students closing their submission merge request
- Removing GitLab resources labels
- Students can modify the protected files CI job

Future Work

- Improve error handling and user feedback
- Process long-running operations asynchronously
- Support additional Version Control Systems (VCS)

Contributions

- Introduces a flexible, automated workflow
- Provides an extensible, open-source, Moodle-native integration
- Reduces manual coordination between LMS and VCS platforms
- Improves consistency and efficiency for instructors and students